WORKED SOLUTION

STEP I 2002, Question 9

Mechanics • Toppling on an Incline

STEP I 2002, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

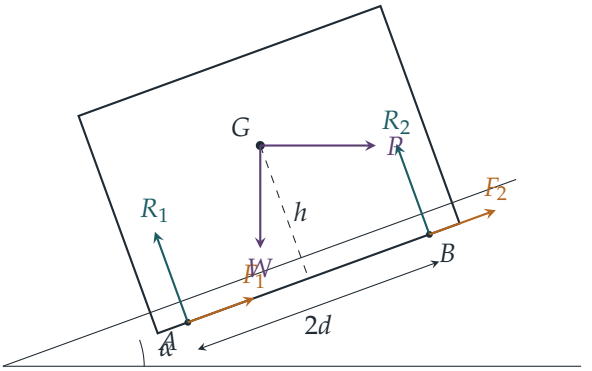
step.maths.org/questions/02-s1-q9

The key idea

The lorry is a rigid body in equilibrium, so we have three equations: resolve along the plane, resolve perpendicular to it, and take moments. Taking moments about G is the clever choice, because both the weight and P pass through G and drop out, leaving only the wheel forces. That one equation links the difference of the normal reactions to the total friction, and it does almost all the work. "Not tipping" then simply means that neither normal reaction is negative.

Solution

We model the lorry as a rigid body whose wheels touch the plane at two points, A (lower) and B (higher), a distance $2d$ apart. Let the normal reactions be R_1 at A and R_2 at B, and let the frictional forces be F_1 and F_2, both taken as positive *up* the plane (either may turn out negative). We assume friction is large enough to prevent sliding; the question is only about tipping. As in the diagram, P is horizontal, pointing towards the slope, so it has a component $P\cos\alpha$ up the plane and a component $P\sin\alpha$ into the plane.



Resolving perpendicular to the plane and along it (up the slope positive):

$$R_1 + R_2 = W\cos\alpha + P\sin\alpha, \tag{1}$$
$$F_1 + F_2 = W\sin\alpha - P\cos\alpha. \tag{2}$$

Now take moments about G. The weight and P act through G, so they contribute nothing. Each normal reaction acts at a perpendicular distance d from G, and each frictional force acts along the plane at a perpendicular distance h from G. Taking anticlockwise as positive (with the plane rising to the right, as in the diagram), R_2 turns the lorry anticlockwise about G, R_1 turns it clockwise, and a friction force up the plane, acting below G, turns it anticlockwise. So

$$d(R_2 - R_1) + h(F_1 + F_2) = 0. \tag{3}$$

Part (i)

If $R_1 = R_2$, equation (3) becomes $h(F_1 + F_2) = 0$. Since $h > 0$, this gives $F_1 + F_2 = 0$: the frictional forces sum to zero, as required.

(From (2), this happens exactly when $P = W\tan\alpha$: the up-slope part of P exactly balances the down-slope part of the weight, so the plane needn't supply any net friction.)

Part (ii)

Substituting (2) into (3),

$$R_2 - R_1 = \frac{h}{d}(P\cos\alpha - W\sin\alpha),$$

and combining this with (1),

$$2R_1 = W\cos\alpha + P\sin\alpha - \frac{h}{d}(P\cos\alpha - W\sin\alpha),$$
$$2R_2 = W\cos\alpha + P\sin\alpha + \frac{h}{d}(P\cos\alpha - W\sin\alpha).$$

Ask yourself What does "does not tip over" mean in terms of these equations?

The ground can push on a wheel but can't pull it, so each normal reaction must satisfy $R \geq 0$. If the formula for R_2 came out negative, the higher wheel would have to be held down; in reality it lifts and the lorry rotates about A. Similarly $R_1 < 0$ means rotation about B. So the lorry stays upright exactly when $R_1 \geq 0$ and $R_2 \geq 0$.

Throughout we use $\frac{d}{h} = \tan\beta$, with $0 < \beta < \frac{\pi}{2}$, and $0 < \alpha < \frac{\pi}{2}$.

The higher wheel stays down ($R_2 \geq 0$). Multiplying $2R_2 \geq 0$ by $d/h > 0$,

$$\tan\beta\,(W\cos\alpha + P\sin\alpha) + P\cos\alpha - W\sin\alpha \geq 0.$$

Multiplying by $\cos\beta > 0$ and collecting terms,

$$P(\cos\alpha\cos\beta + \sin\alpha\sin\beta) \geq W(\sin\alpha\cos\beta - \cos\alpha\sin\beta),$$

that is,

$$P\cos(\alpha-\beta) \geq W\sin(\alpha-\beta).$$

Since $-\frac{\pi}{2} < \alpha - \beta < \frac{\pi}{2}$, we have $\cos(\alpha-\beta) > 0$, so we may divide by it without reversing the inequality:

$$P \geq W\tan(\alpha-\beta).$$

The lower wheel stays down ($R_1 \geq 0$). In the same way, $2R_1 \geq 0$ gives

$$\tan\beta\,(W\cos\alpha + P\sin\alpha) - P\cos\alpha + W\sin\alpha \geq 0,$$

and multiplying by $\cos\beta > 0$,

$$P(\cos\alpha\cos\beta - \sin\alpha\sin\beta) \leq W(\sin\alpha\cos\beta + \cos\alpha\sin\beta),$$

that is,

$$P\cos(\alpha+\beta) \leq W\sin(\alpha+\beta).$$

Provided $\alpha + \beta < \frac{\pi}{2}$ (which the stated answer implicitly assumes, since otherwise $\tan(\alpha+\beta)$ is negative or undefined), $\cos(\alpha+\beta) > 0$, and dividing gives

$$P \leq W\tan(\alpha+\beta).$$

If instead $\alpha + \beta \geq \frac{\pi}{2}$, the left-hand side is ≤ 0 for $P \geq 0$ while the right-hand side is positive, so no horizontal push towards the slope, however large, can tip the lorry over its higher wheel.

Putting the two conditions together, the lorry does not tip if and only if

$$W\tan(\alpha-\beta) \leq P \leq W\tan(\alpha+\beta),$$

which is the required result. Too small a push and it topples down the slope about A; too large and it is pushed over uphill about B.

Checks

- *Geometric picture.* The weight and P combine into a single force through G, at angle θ to the vertical with $\tan\theta = P/W$. The lorry is about to tip when this resultant passes through a wheel contact. The line GA makes angle β with the normal to the plane, and the normal makes angle α with the vertical, so the resultant passes through A when $\theta = \alpha - \beta$, and through B when $\theta = \alpha + \beta$. These are exactly the two end-points.
- *No slope.* With $\alpha = 0$ the condition becomes $|P| \leq Wd/h$: the familiar condition for a box on level ground not to topple under a sideways push at height h.
- *Numbers.* With $W = 3$, $\alpha = 0.4$, $d = 1$, $h = 1.5$, the interval is $-0.571 \leq P \leq 4.551$. Checking $R_1, R_2 \geq 0$ directly against the interval for 20 000 random configurations gave complete agreement, with $R_2 = 0$ at the lower end and $R_1 = 0$ at the upper end.

Common slips

- Taking moments about a wheel instead of G. That works too, but then the weight and P have awkward moment arms, and sign errors creep in.
- Forgetting that the frictional forces have a moment about G. They act at the plane, a distance h below G, and this is where h enters the answer.
- Dividing by $\cos(\alpha \pm \beta)$ without checking its sign, which is what decides the direction of the inequality.

Beyond STEP

Friction between the wheels and the ground is *statically indeterminate*: equilibrium fixes only $F_1 + F_2$, never F_1 and F_2 separately, and finding the split needs a model of how the tyres deform. Engineers meet this all the time in structures with more supports than they strictly need. The geometric check above is the idea of the *line of action of the resultant* and the support polygon, which also underlies the stability of walking robots and stacked blocks. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers the statics of rigid bodies carefully.

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